**Slide 1**

Does everyone remember our definition? Yes?  no, we're not sure! okay let's recap.
genetic modification is when we change the genes inside of a living thing to make it grow in a different way
okay all right. So, you know we're going to talk specifically about ways to make genetically modified food or GM food for short.
Here we go!!!

**Slide 2**

How is GM food produced?

Genetic modification of a particular plant or animal species can be accomplished by a number of different ways.

The DNA alteration of a plant or animal's DNA which holds the genetic information of the species.

In order to change a certain characteristic→ the corresponding gene coding for it, must initially be isolated in order to be inserted into a new DNA strand using a transfer vector.

Then inside the cell the vector replicates and becomes part of the cell's own DNA, altering the organism's characteristics.

Gene-gun method (predominantly used in plant modification) Pellets of metal 'coated with the desirable DNA' are fired at the target cells which are then allowed to reproduce, and may possibly be cloned in order to produce a 'genetically identical crop.'

## Slide 3

- right now we're going to look at three types of genetic modification
- first modifications to make food stay fresh longer
- second modifications to make crops grow better
- and third modifications to make food healthier

## Slide 4

- Our first example is about a modification made to help food retain freshness longer it was called the flavor saver tomato
- And like the name says it was supposed to save the flavor of a fresh tomato
- It's interesting actually because it was the first genetically modified food to be sold in US supermarkets, back in 1994
- So why was this tomato created?
- Well there's a big problem with growing fruits and vegetables and that's keeping them fresh
- Like tomatoes, after you picked them they go bad quickly
- they get too soft ; they don't taste good anymore

**Slide 5**

- Now, the reason the tomato goes bad is because of a special chemical inside the tomato plant
- This chemical starts to work after a tomato is picked it starts making it gets soft
- Scientists wanted to stop this chemical from working
- To do this, they created the flavor saver tomato by adding a gene called antisense RNA
- This antisense RNA gene stops the chemical that makes tomatoes get soft
- The chemical doesn't function so the tomato stays fresh for a long time ; much longer than a normal tomato
- but there were some problems with the flavor saver tomato
- one problem was that people didn't trust this new genetically modified food
- They thought it might be unhealthy or even dangerous to eat
- So, they didn't buy it!
- Another problem was that  no people said it just didn't taste very good
- At any rate, shoppers didn't purchase the flavor saver so it wasn't growing anymore after 1997.

## Slide 6

- All farmers have problems with insects eating their crops. Traditionally, farmers have sprayed their crops with pesticides to kill the insect pests
- But this is **time-consuming**, **expensive**, and often **harmful to the farm workers** in contact with the poisons
- Also, pesticides kill useful insects such as pollinators

it can harm other animals that eat the insects that have been poisoned spraying the pesticides also damage soil and water in the environment

- crops have been genetically modified with a gene from a bacterium: → called **bacillus thuringiensis**; This gene causes the cells of the crop plant to produce an insecticide ; in their leaves that kill insects that eated

- it the genetic modification reduces the need for chemical pesticides; The growth of weeds in fields reduces the yield of crops
- Because the weeds compete with the crops for water light and minerals in the soil.
- Farmers can spray their fields with weed killers, but this may also damage the environment or even the crop plants themselves → So crops can also be genetically modified to be resistant to weed killers.
- Which means that only the weeds will be killed by the weed killers.
- This doesn't reduce the number of chemicals sprayed on a field, but it does mean that higher yields can be gained.

**Slide 7**

- Have you ever been told that eating carrots help you see in the dark?
- This is because carrots are a good source of vitamin a -which is essential for healthy vision
- a common problem in some countries is a deficiency in vitamin a because people have a diet mainly made up of cereals like rice with a few fresh fruits and vegetables
- to overcome this vitamin deficiency rice has been genetically modified to contain beta-carotene which is used in the body to make vitamins this gives the rice a yellow color
- so, it's known as golden rice
- its taste is not altered but it contains extra nutritional benefits